

SOFTWARE TESTING PROJECT

AutomationExercise

Presented By:

Ananya Pattanaik

Emp Id: 85009810

Coforge,

Graduate Engineer Trainee

Testing



Agenda

What we'll cover today

01

Testing Fundamentals

What, Why, When, SDLC & STLC

02

Manual Testing

Test Plan · Scenarios · Cases · Defects · RTM · Peer Review

03

Automation Testing

Selenium POM Framework · Scripts · Reports

04

API & Performance

Postman Collection · JMeter Load Report

05

Project Closing

Challenges · Learnings · Conclusion

What is Software Testing?

Software Testing is the process of evaluating and verifying that a software application or system meets specified requirements and works as expected.

Verification

Are we building the product right?

Validation

Are we building the right product?

Detection

Identify defects before production

Prevention

Reduce rework & technical debt

Why Testing?

Testing Fundamentals

01 **Cost Reduction**

Fixing a bug in production costs 10× more than fixing it during development.

02 **User Satisfaction**

Delivering bug-free software improves user trust and retention.

03 **Security Assurance**

Testing exposes vulnerabilities before attackers do.

04 **Regulatory Compliance**

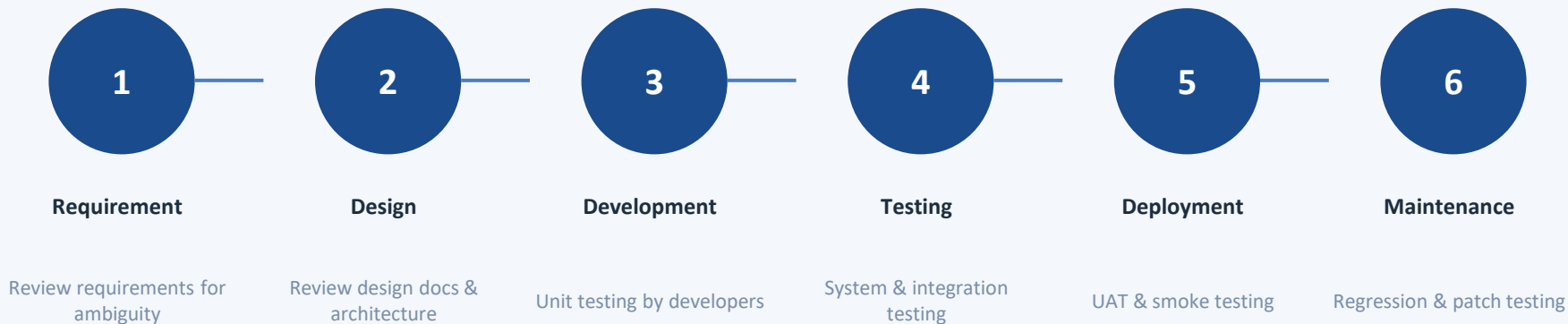
Many industries require formal QA documentation and testing evidence.

05 **Brand Reputation**

A single critical bug can damage brand credibility permanently.

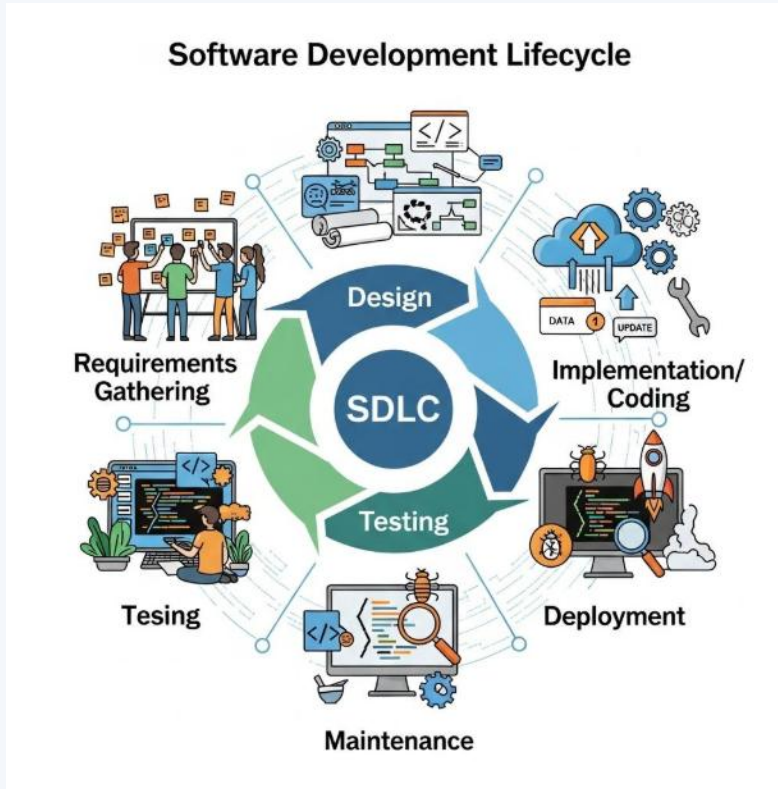
When to Test?

Testing Fundamentals

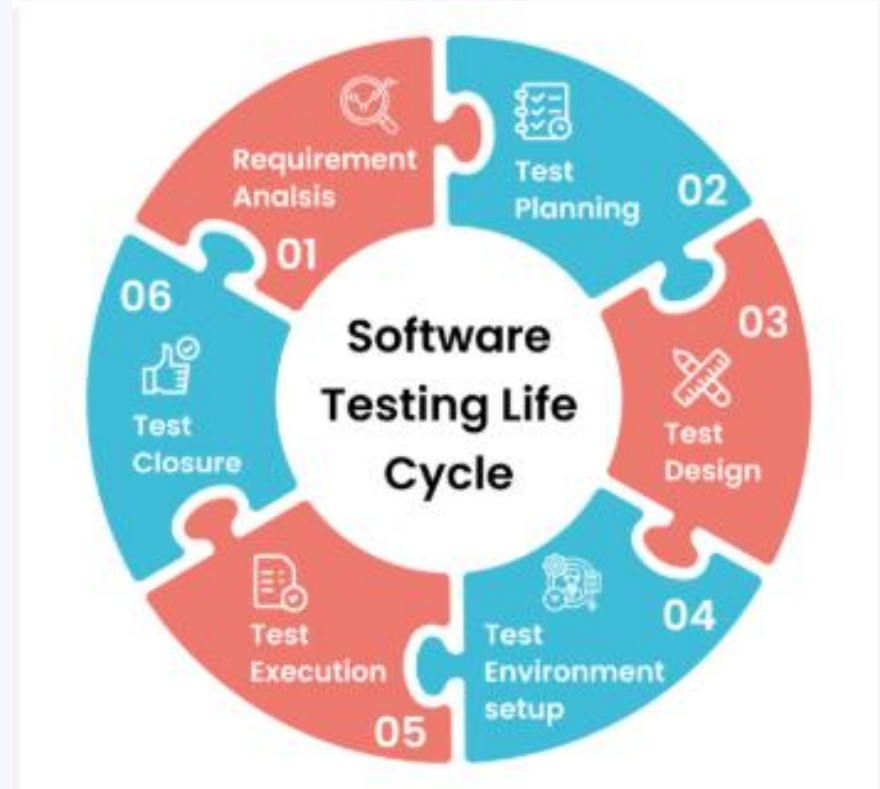


Testing is NOT a phase — it is a continuous activity throughout the SDLC.

Software Development Life Cycle (SDLC)



Software Testing Life Cycle (STLC)



Manual Testing

Manual Testing is a software testing process in which testers manually execute test cases without using automation tools to verify that an application functions according to the specified requirements. It helps identify defects, validate business requirements, and ensure that the software provides a smooth and user-friendly experience. Manual testing is particularly effective for evaluating usability, visual design, and overall application behavior from an end-user's perspective.

For the **AutomationExercise.com** project, manual testing was conducted on various modules such as User Registration, Login and Logout, Product Search, Add to Cart, Checkout Process, and Contact Us. The testing process involved requirement analysis, creation of test scenarios and test cases, execution of test cases, defect reporting, and retesting. This helped ensure the application's functionality, reliability, and quality before proceeding with automation testing.



Life Cycle Of Manual Testing



Test Plan

Test Plan for AutomationExercise Website

1. Test Plan Identifier

File Name: 1.0_AutomationExercise_tplan_v1.0.doc

- Document Name: Test Plan – AutomationExercise Website
- Version: AE_1.0
- Date: 18-June-2026
- Location: Coforge Ltd, Hyderabad

2. Document History

REV	Author	Edits	Date
1.0	Ananya Pattanaik, Graduate Engineer Trainee	Initial Draft	18/06/2026

3. Introduction

AutomationExercise.com is a demo e-commerce web application used for practicing manual and automation testing. It allows users to register, log in, browse products, add items to cart, place orders, and contact support.

This test plan defines the scope, objectives, and approach for testing the functionality, usability, and reliability of the AutomationExercise website.

4. Test Items

Specifies what components/modules will be tested.

Example:

- User Registration Module
- User Login Module
- Product Listing Page
- Product Details Page
- Cart Module
- Checkout Module
- Payment Module
- Contact Us From
- Logout Functionality

5. Features to be Tested

Medium priority features are features that have been stable and High Priority features are features that have been recently changed.

Feature to be Tested	Priority	Comments
User Registration	High	

User Login	High	
Product Search	High	
Product View	High	
Add to Cart	High	
Checkout Process	High	
Contact Us From	High	
Logout	High	

6. Features NOT to be Tested

- Help Contents and user manuals will not be tested.
- Support of the other architectures also will not be tested.
- Payment integration is demo only
- Legacy modules not impacted by current release
- Experimental features

7. Test Environment

LAPTOP-BN0FH82H
Vista by HP Gaming Laptop 15-fabxx

Rename this PC

Device specifications

Copy

Device name

LAPTOP-BN0FH82H

Processor

12th Gen Intel(R) Core(TM) i7-12650H (2.30 GHz)

Installed RAM

16.0 GB (15.7 GB usable)

Device ID

4931345B-00F4-47CF-8973-204E0F6460A3

Product ID

00342-42646-28778-AAOEM

System type

64-bit operating system, x64-based processor

Pen and touch

No pen or touch input is available for this display

Windows specifications

Copy

Edition

Windows 11 Home Single Language

Version

25H2

Installed on

15-12-2024

OS build

26200.7623

Experience

Windows Feature Experience Pack 1000.26100.275.0

Microsoft Services Agreement

Microsoft Software License Terms

8. Test Approach / Test Strategy

Describes how testing will be performed.

The test strategy for AutomationExercise1.0 will include manual testing with automated testing(Selenium With Java) where time/need permits. Automated testing will be used to test the basic Interface objects keyboard and mouse navigation or any redundant tasks best translated to automation. Automated testing will be limited to a Win-64 platform because of Automation tool limitations.

- Sanity test will include an very short battery of tests to ensure the Naukri1.0 runs/launches and can connect to the back end database(MySql /SQL Server 2025).

Test execution will happen as individual component testing. These tests will be fully automated.

- **Acceptance tests** will include a short battery of tests that will only touch major features to ensure AutomationExercise1.0 will operate smoothly at a basic level. These tests will be executed on **Windows 11 Home Single Language**. The goal will be to automate all of these tests.
- Full Functional tests will include a **comprehensive test** of all features listed in this document. These tests will be limited to the Windows 11 pro. These test will include automation with time allowing.

8.1 Types of Testing

- Smoke Testing
- Sanity Testing
- Functional Testing
- Integration Testing
- System Testing
- Regression Testing

8.2 Test Levels

- Unit Testing (by developers)
- Integration Testing
- System Testing
- Acceptance Testing

8.3 Test Techniques

- Black Box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing **Functionality Testing**

Usability Testing

The usability testing will be accomplished by verifying the information in each window are accurate. Menus, icons and toolbar functionality will be tested as applicable to the navigation and results panes. Multi Window Overlapping will be tested because product supports opening of multiple documents.

Robustness/Reliability Testing

Test the GUI for correct keyboard and mouse navigation, windows, menus, buttons, etc. These tests will include Keyboard only navigation of AutomationExercise1.0, Mouse only navigation. Correctness and existence of warning/error dialog boxes. The correct/expected functionality of buttons, icons and menus.

Stress Testing

Not applicable

9. Entry and Exit Criteria

9.1 Entry Criteria

Conditions to start testing:

- Requirements approved
- Test environment ready
- Test data available
- Build deployed

9.2 Exit Criteria

Conditions to stop testing:

- All planned test cases executed

- Critical and high defects closed
- Test summary report prepared

10. Test Deliverables

Test plan	1.0_AE_tplan.doc
Test Specification	1.0_AE_tspec.doc
Test Automation Plan	TBD()
Test Checklist	TBD
Test Logs	TBD
Test tools	Selenium 4.38, Postman, Rest Assured, Jmeter, Jenkins, Git and Git Hub and Appium and MySQL Server 2025

11. Testing Tasks & Resource Needs

- Use of MySQL Server database.
- Write Test Specification.
- Write Test Automation plan.
- Write Test Checklist.
- Setup workspace.
- Develop Automation script.

➤ Minimum requirements for platforms:

12. Schedule

Test Documentation

Document	Development Time		Sign-off Date	Dependencies
	Person-Days	Completion Date		
Test Plan	1 day	18/06/2026	18/06/2026	Functional Spec. Review
Test Specifications	1 day	18/06/2026	18/06/2026	Test plan, Functional Spec, Review
Test Logs	TBD	18/06/2026		
Test Summary Report	TBD	18/06/2026		
Checklist	TBD	18/06/2026		

Test Development

Deliverable	Development Time		Dependencies
	Person-Days	Completion Date	
Functional Test Suite	1	18/06/2026	
Automation build			

Test Execution

Test Activity		Execution Time (Person-Days)			Dependencies
		Alpha	Beta	FCS	
Acceptance		1 Day	1 Day	1Days	
Functional	Functionality	1	1		

<i>Test Activity</i>		<i>Execution Time (Person-Days)</i>			<i>Dependencies</i>
	Usability	All High to Medium test cases			
	Robustness				
	Stress				
	Multi Inst.				
Regression					
Documentation Review		1	1	1	
Total Test Cycle		1	1	1	

13. Risks and Assumptions

Risks:

- **Environment Instability:** AutomationExercise website may be slow or temporarily unavailable, affecting test execution.
- **Internet Dependency:** Testing depends on stable internet connectivity issues may interrupt test runs.
- **Test Dat Limitation:** Limited realistic e-commerce test data may reduce test coverage.

Contingencies:

- **Backup Resources:** Keep alternative internet source to avoid interruptions.
- **Flexible Schedule:** Add buffer time to handle delays due to site or network issues.
- **Prepared Test Data:** Maintain multiple test data sets to ensure smooth testing.

14. Approvals \Sign of

POSITION	NAME	SIGNATURE	REVIEWED	APPROVAL
* Approval required Project Manager				
Project Manager			Raghavendra BN	Raghavendra BN
Client Approval	Coforge			

Test Scenarios

Name: Ananya Pattanaik
Email ID: ananya.m1430@gmail.com
Emp ID: 85009810

TEST SCENARIOS FOR AUTOMATIONEXERCISE.COM

Scenario	Scenario Name	Scenario Description	Preconditions	Expected Outcome
TS_01	Verify user registration	Validate that a new user can register using valid details	User is not registered	User account is created successfully
TS_02	Verify user login	Validate that a registered user can log in with valid credentials	User must be registered	User is logged in and redirected to dashboard
TS_03	Verify invalid login	Validate login with invalid credentials	User enters wrong username/password	Error message is displayed
TS_04	Verify product search	Validate that user can search for products using keywords	User is on homepage	Relevant products are displayed
TS_05	Verify product view	Validate user can view product details	User selects a product	Product details page is displayed
TS_06	Verify add to cart	Validate user can add product to cart	User is on product page	Product is added to cart successfully
TS_07	Verify remove from cart	Validate user can remove product from cart	Product is present in cart	Product is removed successfully
TS_08	Verify checkout process	Validate user can proceed to checkout	Product is in cart	Checkout page is displayed
TS_09	Verify contact us form	Validate user can submit contact form	User is on Contact Us page	Form is submitted successfully
TS_10	Verify logout functionality	Validate that user can logout	User is logged in	User is logged out successfully

Test Cases

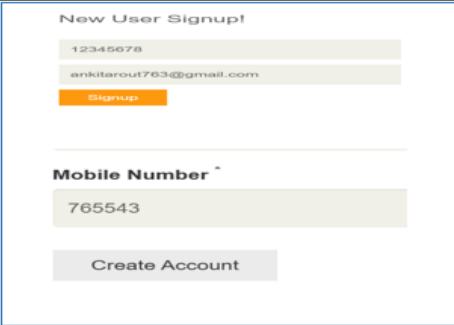
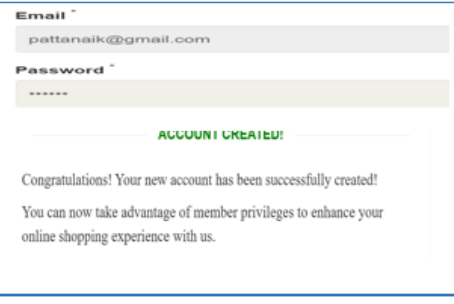
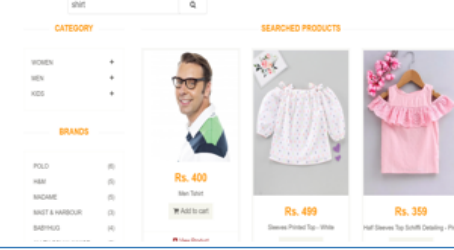
Project Name:	automationexercise.com						
Module Name:	User Login & Logout	Date Updated:	19/06/2026				
Last Build Tested:		Reviewed By:	Raghavendra BN				
Run No:	10 Test Cases	Executed By:	Ananya Pattanaik				
Run Date:	18/06/2026						
Test Objectives:	Mapping Test Scenarios with Test Cases (automationexercise.com)						
Test Setup:	MIS Excel Sheet (Online)						
Req. Id	Test Case ID	Test Case Description	Test steps	Test Data	Expected Result	Actual Result	Status
Req_1.0	TC_01	Verify user can register with valid details	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. Verify that homepage is displayed successfully 4. User should click on Signup button 5. User should enter valid Name and Email address in signup section and click on Signup button 6. User should enter all required details such as Password, Date of Birth, Address, State, City, and Mobile number 7. User should click on 'Create Account' button 8. Verify account creation confirmation message	URL: https://automationexercise.com Name: Test User Email: testuser@gmail.com Password: Test@123	User account should be created successfully and confirmation message should be displayed	User account created successfully and confirmation displayed	Passed
Req_1.1	TC_02	Verify user can login with valid credentials	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should enter valid Email ID and Password 5. User should click on Login button	Email: testuser@gmail.com Password: Test@123	User should be logged in successfully and homepage/dashboard should be displayed	User logged in successfully	Passed

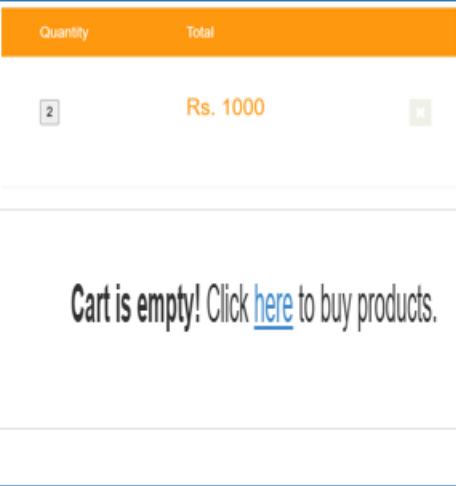
Req_1.2	TC_03	Verify login with invalid credentials	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should enter invalid Email ID and Password 5. User should click on Login button	Email: testuser@gmail.com Password: Test@123	Error message "Invalid credentials" should be displayed	Error message displayed	Passed
Req_1.3	TC_04	Verify product search functionality	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should locate search bar 5. User should enter product keyword (e.g., "Dress") 6. User should Click Search button 7. User should Verify results	Email: testuser@gmail.com Password: Test@123 Keyword: Dress	Relevant products based on keyword should be displayed	Products displayed correctly	Passed
Req_1.4	TC_05	Verify product details page	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should Navigate to product listing 5. User should Click on any product 6. User should Verify product name, price, and details displayed	Email: testuser@gmail.com Password: Test@123 Product: Any available product	Product details page should be displayed with complete info	Product details displayed successfully	Passed

Req_1.5	TC_06	Verify adding product to cart	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should locate search bar 5. User should enter product keyword (e.g., "Dress") 6. User should Click Search button 7. User should Verify results 8. User should Click on 'Add to Cart' button 9. User should View cart	Email: testuser@gmail.com Password: Test@123 Keyword: Dress Product: Dress	Product should be added to cart successfully	Product added successfully	Passed
Req_1.6	TC_07	Verify removing product from cart	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should be in Cart Page 4. User should Click on remove/delete option 5. User should Verify cart	Email: testuser@gmail.com Password: Test@123 Keyword: Dress Product should be added in cart	Product should be removed from cart	Product removed successfully	Passed
Req_1.7	TC_08	Verify checkout process	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should locate search bar 5. User should enter product keyword (e.g., "Dress") 6. User should Click Search button 7. User should Verify results 8. User should Click on 'Add to Cart' button 9. User should be on cart page 10. User should Click on 'Proceed to Checkout' 11. User should Verify checkout page loads	Email: testuser@gmail.com Password: Test@123 Keyword: Dress Product should be added in cart	User should be navigated to checkout page	Checkout page displayed successfully	Passed

Req_1.8	TC_09	Verify contact form submission	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should Navigate to Contact Us page 5. User should Enter Name, Email, Subject, Message 6. User should Click Submit 7. User should Accept alert if displayed	Name: Test User Email: test@gmail.com Message: Success! Your details have been submitted successfully.	Form should be submitted and success message displayed	Form submitted successfully	Passed
Req_1.9	TC_10	Verify user can logout successfully	1. User should open Microsoft Edge browser 2. User should enter URL: https://automationexercise.com 3. User should click on Login option on the Homepage. 4. User should Click on Logout button 5. User should Verify logout	Email: testuser@gmail.com Password: Test@123	User should be logged out and redirected to login/homepage	User logged out successfully	Passed

Defect Report

Defect Description	Steps to Reproduce	Priority	Severity	Status	Created By	Assigned To	Screenshot
Registration form accepts invalid input such as numbers in Name field and less than 10 digits in Mobile number field	1. User should open Microsoft Edge browser 2. Launch https://automationexercise.com 3. Click on 'Signup / Login' 4. Enter Name as numeric value 5. Enter Email and click Signup 6. Fill remaining details 7. Enter Mobile Number with less than 10 digits 8. Click on 'Create Account' 9. Observe system behavior	High	Major	New	Ananya Pattanaik	Raghavendra BN	
Password field accepts weak password without validation	1. User should open Microsoft Edge browser 2. Launch https://automationexercise.com 3. Click on Signup 4. Enter Name and Email and click Signup 5. Enter a weak password such as "12345" or "abc" 6. Fill remaining required details and Click on Create Account 7. Observe system behavior	High	Major	New	Ananya Pattanaik	Raghavendra BN	
Product search shows irrelevant results	1. User should open Microsoft Edge browser 2. Launch https://automationexercise.com 3. User should Click on Products 4. User should Enter keyword like "Shirt" 5. Click search 6. Observe results	High	Critical	New	Ananya Pattanaik	Raghavendra BN	

Cart page does not clearly display remove/delete option, causing confusion in user interaction	1. User should open Microsoft Edge browser 2. Launch https://automationexercise.com 3. User should Add product to cart 4. User should Navigate to cart page 5. Try to find remove/delete option 6. Observe cart behavior 7.Remove product 8.Observe that cart becomes empty but remove option is not clearly visible	Medium	Minor	New	Ananya Pattanaik	Raghavendra BN	
No confirmation message displayed after successful logout	1. User should open Microsoft Edge browser 2. Launch https://automationexercise.com 3. User should be logged in 4. User Clicked on Logout button 5. Observe system behavior	Low	Minor	New	Ananya Pattanaik	Raghavendra BN	

Requirement Traceability Matrix

Req No.	Requirement Description	Test Case ID	Status
R_1.0	The system should allow a new user of AutomationExercise.com to register successfully using valid name, email ID, password, and personal details.	TC_01	TC_01 – Passed
R_1.1	The system should allow a registered user of AutomationExercise.com Login functionality to log in successfully using valid credentials..	TC_02	TC_02 – Passed
R_1.2	The system should display an appropriate error message when a user attempts to log in with invalid credentials in AutomationExercise.com	TC_03	TC_03 – Passed
R_1.3	The system should allow the user of AutomationExercise.com to search for products using keywords in the search bar.	TC_04	TC_04 – Passed
R_1.4	The system should allow the user of AutomationExercise.com to view product details by selecting any product from the product listing page.	TC_05	TC_05 – Passed
R_1.5	The system should allow the user of AutomationExercise.com to add products to the shopping cart from the product listing or details page.	TC_06	TC_06 – Passed
R_1.6	The system should allow the user of AutomationExercise.com to remove products from the shopping cart successfully.	TC_07	TC_07 – Passed
R_1.7	The system should allow the user of AutomationExercise.com to proceed to checkout after adding products to the cart.	TC_08	TC_08 – Passed
R_1.8	The system should allow the user of AutomationExercise.com to submit the contact form with valid details successfully.	TC_09	TC_09 – Passed
R_1.9	The system should allow the user of AutomationExercise.com to log out successfully and redirect to the login/home page.	TC_10	TC_10 – Passed

Peer Review



Ananya Pattanaik <ananya.m1430@gmail.com>

to Raghavendra ▾

Hi Sir,

I have shared the test deliverables for peer review. Please find attached the Test Plan, Test Scenarios, Test Cases, Defect Report and RTM.

Kindly review and share your feedback or suggestions at your convenience.

Thank you for your time and support.

Best Regards,
Ananya Pattanaik



Raghavendra Bn

to me ▾

10:17 AM (10 hours ago)



Hi Ananya,

Thank you very much for sending across the test deliverables. I have successfully received all the documents, including the Test Plan, Test Scenarios, Test Cases, Defect Report, and the Requirements Traceability Matrix (RTM). I appreciate the effort and attention to detail that has gone into preparing these materials.

I will thoroughly reviewed each of the documents to ensure that all aspects of the testing process have been adequately covered. My review will focus on the completeness and accuracy of the test cases, the clarity of the test scenarios, and the traceability between requirements and test cases as documented in the RTM. Additionally, I will examine the defect report to understand the issues identified so far and how they have been addressed.

If there are any particular sections or areas within these documents that you would like me to pay special attention to, please let me know. Your input will help me provide more targeted and constructive feedback.

I have completed my review, I will compile my observations and share detailed feedback with you. If I have any questions or require further clarification during the review process, I will reach out accordingly.

Thank you again for your hard work and for keeping me in the loop. Looking forward to collaborating further to ensure the quality and success of this project.

Good To Go

Best regards,
Raghavendra BN

Automation Testing

Automation Testing is a software testing technique in which test cases are executed automatically using testing tools and scripts. It helps verify the functionality of an application efficiently by reducing manual effort and increasing test execution speed. Automation testing is particularly useful for repetitive tasks, regression testing, and validating large applications with high accuracy and consistency.

For the **AutomationExercise.com** project, automation testing was implemented using **Selenium WebDriver, Java, TestNG, and Maven**. Automated scripts were developed for key functionalities such as User Registration, Login, Product Search, Add to Cart, and Checkout. This approach improved testing efficiency, reduced execution time, and ensured reliable validation of the application's features across multiple test runs.



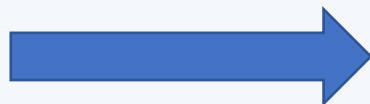
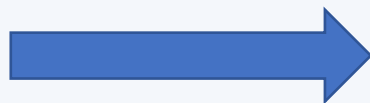
Tools & Technology Stack

JDK(JAVA)	21(Latest Stable)
Selenium	4.39.0
TestNG	7.11
Apache POI	5.5.1
Postman	11.70.0
Eclipse	2022

Allure Reports	2.17.0
Apache JMeter	5.6.3
MySQL Workbench	8.0.34
Jenkins	LTS 2.414.3
JDBC	

Folder Structure

- AutomationExercise_Project
 - src/main/java
 - src/main/resources
 - src/test/java
 - com.Framework.Pages
 - com.Framework.TestCase
 - utility
 - src/test/resources
 - JRE System Library [JavaSE-1.8]
 - Maven Dependencies
 - allure-results
 - Configuration
 - Screenshots
 - src
 - target
 - TestData
 - test-output
 - pom.xml



- src/test/java
 - com.Framework.Pages
 - BaseClass.java
 - LoginPage.java
 - com.Framework.TestCase
 - LoginAutomationExercise.java
 - utility
 - BrowseFactory.java
 - ConfigDataProvider.java
 - ExcelDataProvider.java
 - Helper.java
- Configuration
 - config.properties
- Screenshots
 - AutomationExercise_06_20_2026_11_47_41.png
 - AutomationExercise_06_20_2026_11_49_52.png
- src
- target
- TestData
 - LoginCredential.xlsx

POM.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="https://www.w3.org/2001/XMLSchema-  
  <modelVersion>4.0.0</modelVersion>  
  <groupId>AutomationExercise_Project</groupId>  
  <artifactId>AutomationExercise_Project</artifactId>  
  <version>0.0.1-SNAPSHOT</version>  
  
  <dependencies>  
    <!-- Source: https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->  
    <dependency>  
      <groupId>org.seleniumhq.selenium</groupId>  
      <artifactId>selenium-java</artifactId>  
      <version>4.44.0</version>  
      <scope>compile</scope>  
    </dependency>  
    <!-- Source: https://mvnrepository.com/artifact/org.testng/testng -->  
    <dependency>  
      <groupId>org.testng</groupId>  
      <artifactId>testng</artifactId>  
      <version>7.12.0</version>  
      <scope>test</scope>  
    </dependency>  
    <!-- Source: https://mvnrepository.com/artifact/au.com.bytecode/opencv -->  
    <dependency>  
      <groupId>au.com.bytecode</groupId>  
      <artifactId>opencv</artifactId>  
      <version>2.4</version>  
      <scope>compile</scope>  
    </dependency>
```

```
    <groupId>org.apache.poi</groupId>  
    <artifactId>poi</artifactId>  
    <version>5.5.1</version>  
    <scope>compile</scope>  
  </dependency>  
  
  <!-- Source: https://mvnrepository.com/artifact/org.apache.poi/poi-ooxml -->  
  <dependency>  
    <groupId>org.apache.poi</groupId>  
    <artifactId>poi-ooxml</artifactId>  
    <version>5.5.1</version>  
    <scope>compile</scope>  
  </dependency>  
  
  <!-- Source: https://mvnrepository.com/artifact/io.qameta.allure/allure-testng -->  
  <dependency>  
    <groupId>io.qameta.allure</groupId>  
    <artifactId>allure-testng</artifactId>  
    <version>2.35.2</version>  
    <scope>compile</scope>  
  </dependency>  
  
  <!-- Source: https://mvnrepository.com/artifact/org.apache.maven.plugins/maven-surefire-report-plugin -->  
  <dependency>  
    <groupId>org.apache.maven.plugins</groupId>  
    <artifactId>maven-surefire-report-plugin</artifactId>  
    <version>3.5.6</version>  
    <scope>compile</scope>  
  </dependency>
```


Helper.java

```
package utility;

import org.openqa.selenium.WebDriver;

public class Helper {

    public static void capturedScreenShot(WebDriver driver) {

        try {
            TakesScreenshot ts = (TakesScreenshot) driver;
            File source = ts.getScreenshotAs(OutputType.FILE);
            FileUtils.copyFile(source, new File("./ScreenShots/AutomationExercise_" + getCurrentDateTime() + ".png"));
            System.out.println("Captured ScreenShot - On Failure");
        } catch (Exception e) {
            System.out.println("Exception While Taking ScreenShot " + e.getMessage());
        }
    }

    public static String getCurrentDateTime() {
        DateFormat customFormat = new SimpleDateFormat("MM_dd_yyyy_HH_mm_ss");
        Date currentdate = new Date();
        return customFormat.format(currentdate);
    }
}
```

BrowseFactory.java

```
package utility;

import org.openqa.selenium.WebDriver;

public class BrowseFactory {
    WebDriver driver;

    public static WebDriver BrowserOptions(WebDriver driver, String browserName, String AppUrl) {
        if (browserName.equalsIgnoreCase("chrome")) {
            driver = new ChromeDriver();
        } else if (browserName.equalsIgnoreCase("firefox")) {
            driver = new FirefoxDriver();
        } else if (browserName.equalsIgnoreCase("edge")) {
            driver = new EdgeDriver();
        } else {
            System.out.println("Browser Does Not Support....");
        }

        driver.manage().window().maximize();
        driver.get(AppUrl);
        return driver;
    }

    public static void quitBrowser(WebDriver driver) {
        driver.quit();
    }
}
```

ConfigDataProvider.java

```
package utility;

import java.util.Properties;

public class ConfigDataProvider {
    Properties pro;
    public ConfigDataProvider() {
        File src = new File("../Configuration/config.properties");
        try {
            FileInputStream fis = new FileInputStream(src);
            pro = new Properties();
            pro.load(fis);
        } catch (Exception e) {
            System.out.println("Unable to load properties file " + e.getMessage());
        }
    }

    public String getBrowser() {
        return pro.getProperty("Browser");
    }

    public String getAppURL() {
        return pro.getProperty("AppUrl");
    }
}
```

ExcelDataProvider.java

```
package utility;

import java.io.FileInputStream;

public class ExcelDataProvider {
    XSSFWorkbook wb;
    public ExcelDataProvider() {
        try {
            String path = System.getProperty("user.dir") + "/TestData/LoginCredential.xlsx";
            FileInputStream fis = new FileInputStream(path);
            wb = new XSSFWorkbook(fis);
        } catch (Exception e) {
            System.out.println("Unable to load Excel file " + e.getMessage());
        }
    }

    public String getCellData(String sheetName, int row, int column) {
        return wb.getSheet(sheetName)
            .getRow(row)
            .getCell(column)
            .toString(); // Works for your Excel
    }
}
```

BaseClass.java

```
package com.Framework.Pages;
import org.openqa.selenium.WebDriver;

public class BaseClass {
    public WebDriver driver;
    public utility.ExcelDataProvider excel;
    public ConfigDataProvider config;

    @BeforeSuite
    public void SetUp() {
        excel = new ExcelDataProvider();
        config = new ConfigDataProvider();
    }

    @BeforeClass
    public void BrowserTest() {
        driver = BrowseFactory.BrowserOptions(driver, config.getBrowser(), config.getAppURL());
    }

    @AfterClass
    public void tearDown() {
        BrowseFactory.quitBrowser(driver);
    }

    @AfterMethod
    public void tearDownMethod(ITestResult result) {
        if (result.getStatus() == ITestResult.FAILURE) {
            Helper.capturedScreenShot(driver);
        }
    }
}
```

LoginPage.java

```
package com.Framework.Pages;
import org.openqa.selenium.WebDriver;

public class LoginPage {

    WebDriver driver;

    public LoginPage(WebDriver driver) {
        this.driver = driver;
        PageFactory.initElements(driver, this);
    }

    @FindBy(xpath = "//input[@data-qa='login-email']")
    WebElement loginEmail;
    @FindBy(xpath = "//input[@data-qa='login-password']")
    WebElement loginPassword;
    @FindBy(xpath = "//button[@data-qa='login-button']")
    WebElement loginButton;

    public void loginApp(String email, String password) {

        loginEmail.sendKeys(email);
        loginPassword.sendKeys(password);
        loginButton.click();
    }
}
```

LoginAutomationExercise.java

```
package com.Framework.TestCase;
```

```
+ import org.openqa.selenium.support.PageFactory; ..
```

Run ALL

```
public class LoginAutomationExercise extends BaseClass {
```

@Test

Run | Debug

```
- public void loginApp() throws Exception {
```

```
    ExcelDataProvider excel = new ExcelDataProvider();
```

```
    LoginPage page = PageFactory.initElements(driver, LoginPage.class);
```

```
    page.loginApp(
```

```
        excel.getCellData("Sheet1", 0, 0),
```

```
        excel.getCellData("Sheet1", 0, 1) |
```

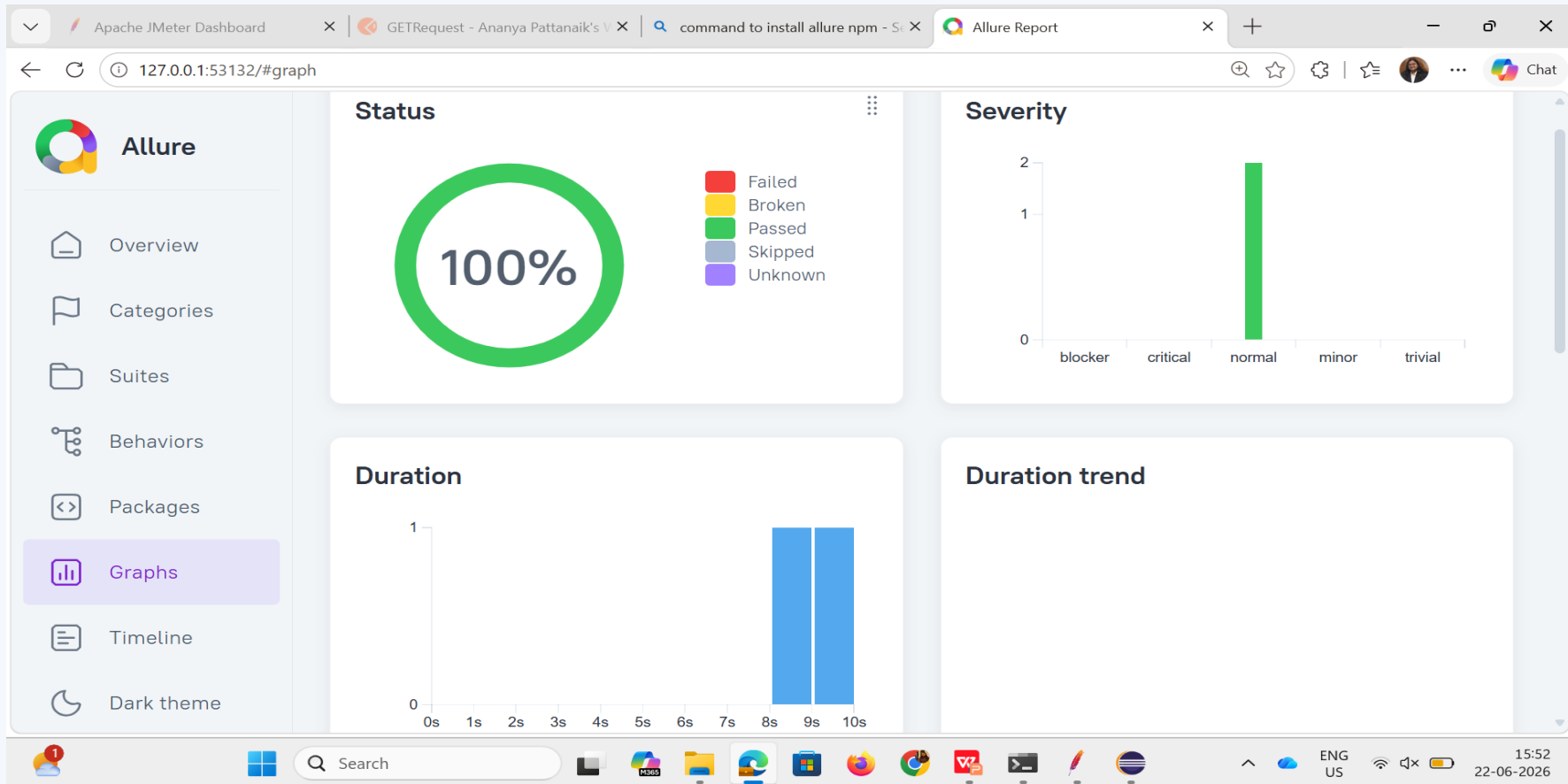
```
    );
```

```
    Thread.sleep(3000);
```

```
}
```

```
}
```

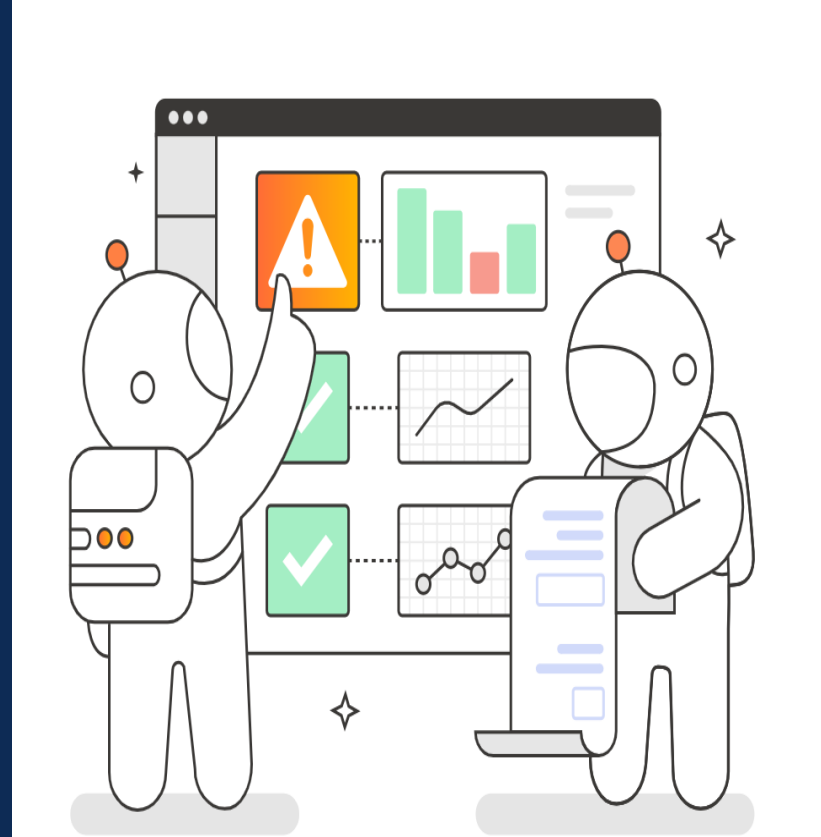
Allure Report



API & Performance Testing

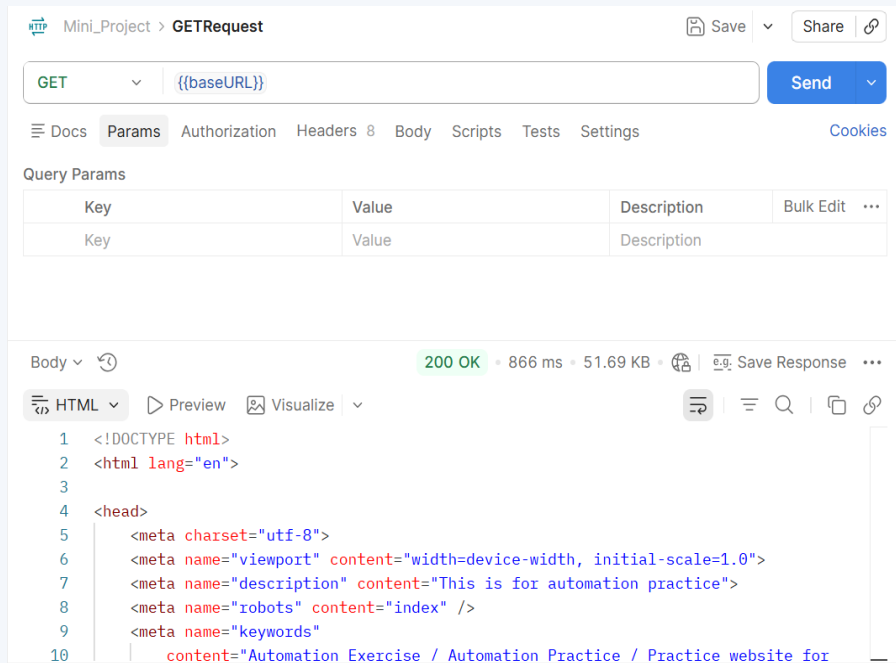
API testing is a type of software testing that verifies the functionality, reliability, and performance of application programming interfaces (APIs). It involves sending requests to API endpoints and validating responses to ensure correct data exchange between systems. API testing focuses on request and response validation, status codes, and error handling using methods like GET, POST, PUT, and DELETE.

It is important because APIs act as the backbone of modern applications, enabling communication between different components. API testing helps detect defects early, ensures smooth integration, and improves overall system performance. Tools like Postman and JMeter are commonly used to perform and automate API testing, making applications more stable and secure.



Postman

GET Request



The screenshot shows the Postman interface for a GET request. The URL bar contains `{{baseUrl}}`. The response status is 200 OK, with a response time of 866 ms and a body size of 51.69 KB. The response body is displayed in HTML format, showing a basic HTML document structure with a head section containing meta tags for charset, viewport, description, robots, and keywords.

Mini_Project > GETRequest

GET `{{baseUrl}}` Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

Query Params

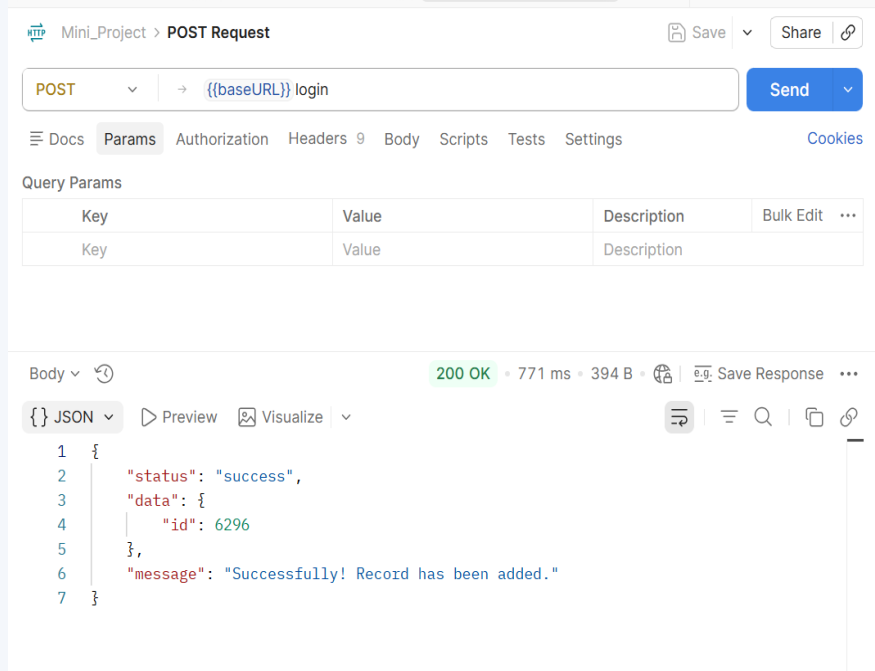
Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `<>` 200 OK • 866 ms • 51.69 KB • Save Response ...

HTML Preview Visualize

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   <meta name="description" content="This is for automation practice">
8   <meta name="robots" content="index" />
9   <meta name="keywords"
10     content="Automation Exercise / Automation Practice / Practice website for
```

POST Request



The screenshot shows the Postman interface for a POST request. The URL bar contains `{{baseUrl}}` login. The response status is 200 OK, with a response time of 771 ms and a body size of 394 B. The response body is displayed in JSON format, showing a success message and a data object with an id of 6296.

Mini_Project > POST Request

POST `{{baseUrl}}` login Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body `<>` 200 OK • 771 ms • 394 B • Save Response ...

{ } JSON Preview Visualize

```
1 {
2   "status": "success",
3   "data": {
4     "id": 6296
5   },
6   "message": "Successfully! Record has been added."
7 }
```

JMeter

AutomationExercise.jmx (C:\Users\Ananya.Pattanaik\OneDrive - Coforge Limited\Desktop\FinalProject\JmeterTesting\AutomationExercise.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:00 0 0/0

Test Plan

- AutomationExercise
 - HomePage Request
 - Constant Timer
 - Products Request
 - Constant Timer
 - Cart Request
 - Constant Timer
 - Login Request
 - Constant Timer
 - ContactUs Request
 - Constant Timer
 - Mast&Harbour Request
 - Constant Timer
 - Madame Request
 - Constant Timer
 - KookieKids Request
 - Constant Timer
 - BabyHug Request
 - Constant Timer
 - BIBA Request
 - Constant Timer
 - View Results Tree
 - Aggregate Graph
 - Random Controller

Thread Group

Name: AutomationExercise

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 10

Ramp-up period (seconds): 1

Loop Count: ☐ Infinite 2

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

Duration (seconds):

Startup delay (seconds):

Search



34°



Search



ENG
US



12:39
22-06-2026

Aggregate Graph

AutomationExercise.jmx (C:\Users\Ananya.Pattanaik\OneDrive - Coforge Limited\Desktop\FinalProject\JmeterTesting\AutomationExercise.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help



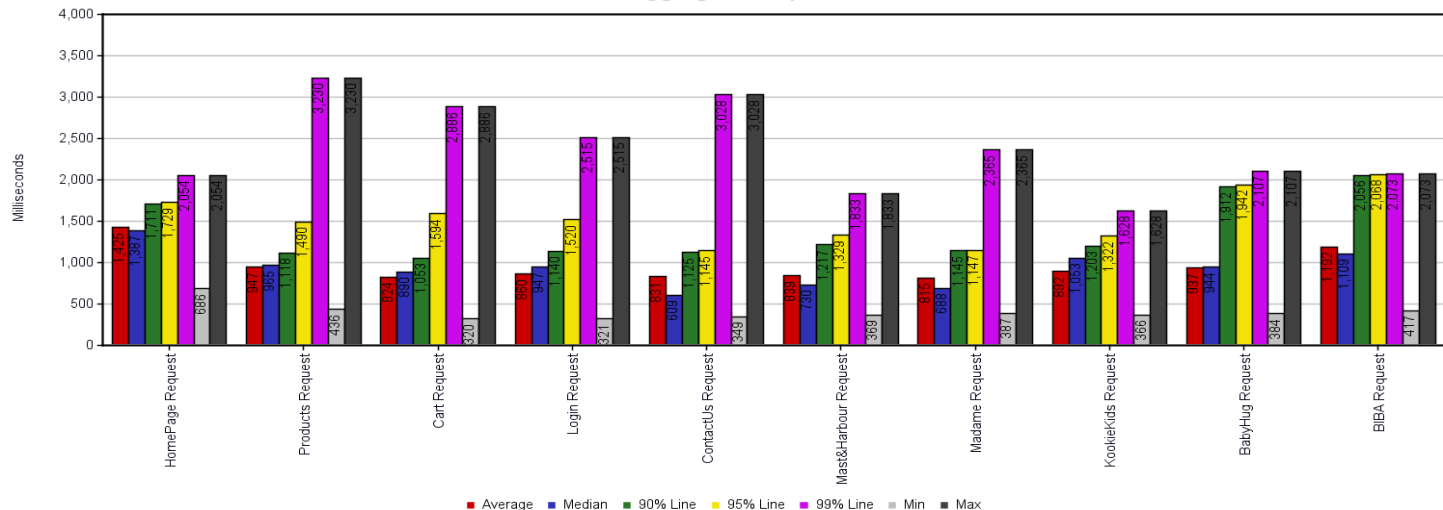
00:02:09 0 0/10

- Test Plan
 - AutomationExercise
 - HomePage Request
 - Constant Timer
 - Products Request
 - Constant Timer
 - Cart Request
 - Constant Timer
 - Login Request
 - Constant Timer
 - ContactUs Request
 - Constant Timer
 - Mast&Harbour Request
 - Constant Timer
 - Madame Request
 - Constant Timer
 - KookieKids Request
 - Constant Timer
 - BabyHug Request
 - Constant Timer
 - BIBA Request
 - Constant Timer
 - View Results Tree
 - Aggregate Graph
 - Random Controller

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB...	Sent
HomePage ...	20	1425	1387	1711	1729	2054	686	2054	0.00%	18.2/min	15.82	
Products Re...	20	947	965	1118	1490	3230	436	3230	0.00%	18.2/min	14.58	
Cart Request	20	824	890	1053	1594	2886	320	2886	0.00%	17.6/min	2.42	
Login Reau	20	960	947	1140	1520	2515	321	2515	0.00%	17.5/min	2.22	

Settings Graph

Aggregate Graph



32°



Search



ENG
US

12:47
22-06-2026

Perf / 35

HTML Report

```
C:\ApacheJmeterInstallation\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>jmeter -n -t "C:\Users\Ananya.Pattanaik\OneDrive - Coforge Limited\Desktop\FinalProject\JmeterTesting\AutomationExercise.jmx" -l "C:\Users\Ananya.Pattanaik\OneDrive - Coforge Limited\Desktop\FinalProject\JmeterTesting\NONGUI\TestFile\AutomationExercise.csv" -e -o "C:\Users\Ananya.Pattanaik\OneDrive - Coforge Limited\Desktop\FinalProject\JmeterTesting\HTML\Reports"
```

Apache JMeter Dashboard

[Dashboard](#)

[Charts](#)

[Customs Graphs](#)

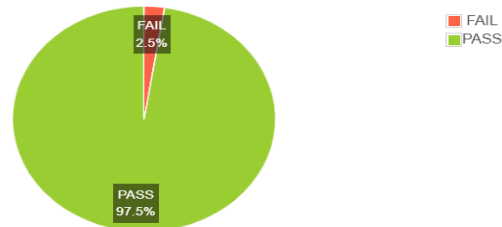
Test and Report information

Source file	"AutomationExercise.csv"
Start Time	"21/06/26, 12:16 am"
End Time	"21/06/26, 12:18 am"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.588	500 ms	1 sec 500 ms	Total
0.425	500 ms	1 sec 500 ms	HomePage Request
0.450	500 ms	1 sec 500 ms	Products Request
0.600	500 ms	1 sec 500 ms	Cart Request
0.600	500 ms	1 sec 500 ms	Login Request
0.600	500 ms	1 sec 500 ms	KookieKids Request

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	200	5	2.50%	826.27	137	1916	899.00	1232.30	1558.45	1809.99	1.73	34.29	0.24
BabyHug Request	20	0	0.00%	707.70	304	1076	867.50	1047.70	1074.60	1076.00	0.32	4.96	0.05
BIBA Request	20	1	5.00%	689.55	319	1003	864.00	940.60	999.90	1003.00	0.32	5.02	0.04
Cart Request	20	0	0.00%	823.75	282	1485	883.50	1133.40	1468.10	1485.00	0.33	2.68	0.04
ContactUs Request	20	0	0.00%	776.95	289	1748	876.50	1110.10	1716.30	1748.00	0.33	3.20	0.04
HomePage Request	20	0	0.00%	1207.15	340	1916	1352.00	1809.90	1910.70	1916.00	0.32	16.73	0.04
KookieKids Request	20	1	5.00%	796.45	290	1109	921.00	1105.70	1108.85	1109.00	0.32	4.42	0.05
Login Request	20	0	0.00%	777.45	282	984	883.00	965.20	983.20	984.00	0.33	2.50	0.04
Madame Request	20	1	5.00%	678.25	137	1273	694.00	1212.80	1271.40	1273.00	0.32	5.09	0.05
Mast&Harbour Request	20	1	5.00%	774.25	307	1073	907.50	993.40	1069.20	1073.00	0.32	4.44	0.05
Products Request	20	1	5.00%	1031.20	341	1561	989.00	1504.40	1558.45	1561.00	0.33	14.84	0.04

RestAssured

GET Request

```
1 package com.restAssured.Demo;
2 import static io.restassured.RestAssured.*;
3
4 public class BDDGetRequest3 {
5
6     public static void main(String[] args) {
7         Response response = given().when().get("https://dummy.restapiexample.com/api/v1/employee/1").then().extract().response();
8         System.out.println("Status code " + response.getStatusCode());
9
10        System.out.println("complete response: ");
11        System.out.println(response.asPrettyString());
12
13        System.out.println("Response body: ");
14        System.out.println(response.getBody().asPrettyString());
15
16        System.out.println("Status line: " + response.getStatusLine());
17
18        System.out.println("Content Type " + response.getContentType());
19
20        System.out.println("Response Time " + response.getTime() + "ms");
21    }
22 }
```

Problems Javadoc Declaration Console X Results of running method CompletePutPostTest.testPostRequestComplete

```
:terminated> BDDGetRequest3 [Java Application] C:\Users\Ankita.Rout\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\
Status code 200
complete response:
{
  "status": "success",
  "data": {
    "id": 1,
    "employee_name": "Tiger Nixon",
    "employee_salary": "320800".
  }
}
```

POST Request

```
1 package com.example.tests;
2 import io.restassured.RestAssured;
10
11 Run ALL
12 public class CompletePutPostTest {
13     private static final String BASE_URL = "https://jsonplaceholder.typicode.com";
14     private JSONObject testUser;
15     private int createdUserId;
16
17     @BeforeClass
18     public void setup() {
19         RestAssured.baseURI = BASE_URL;
20         // Common test data
21         testUser = new JSONObject();
22         testUser.put("name", "Test User");
23         testUser.put("email", "raghu.asteapahead@gmail.com");
24         testUser.put("username", "raghavendra");
25     }
26     @Test(priority = 1)
27     Run / Debug
28     public void testPostRequestComplete() {
29         System.out.println("\n\n===== POST REQUEST - TRADITIONAL =====");
```

Problems @ Javadoc Declaration Console Results of running class CompletePutPostTest

```
terminated> CompletePutPostTest [TestNG] C:\Users\Ankita.Rout\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64.jre\bin\java.exe
-- POST Response Details --
. Status Code: 201
. Complete Response: {
  "name": "Test User",
  "id": 11,
  "email": "raghu.asteapahead@gmail.com",
  "username": "raghavendra"
}
. Response Body: {
```

PUT Request

```
Run / Debug
80 public void testPutRequestBDDComplete() {
81     System.out.println("\n\n===== PUT REQUEST - BDD =====");
82     JSONObject updatedPost = new JSONObject();
83     updatedPost.put("id", 1);
84     updatedPost.put("title", "BDD Updated Post");
85     updatedPost.put("body", "Updated using BDD approach");
86     updatedPost.put("userId", 1);
87
88     System.out.println("Request Body: " + updatedPost.toString(2));
89
90     given().header("Content-Type", "application/json")
91         .body(updatedPost.toString())
92         .when().put("/posts/1")
93         .then()
94             .statusCode(200)
95             .statusCode(containsString("200"))
96             .contentType("application/json")
97             .body("title", equalTo("BDD Updated Post"))
98             .body("body", equalTo("Updated using BDD approach"))
99             .body("userId", equalTo(1))
100             .time(lessThan(5000L))

```

Problems Javadoc Declaration Console Results of running class CompletePutPostTest

```
terminated> CompletePutPostTest [TestNG] C:\Users\Ankita.Rout\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.f
===== PUT REQUEST - TRADITIONAL =====
Request Body: {
  "phone": "1-555-555-9999",
  "name": "Updated Test User",
  "id": 1,
  "email": "raghu.astepahead@example.com",
  "username": "updateduser"
}
```

Cucumber

The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure for 'cucumber.first.example', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'JRE System Library [jdk-21.0.2]', 'Maven Dependencies', 'Futures', 'LoginTest.feature', 'TestCase2.feature', 'testCase3.feature', 'src', 'target', and 'pom.xml'.
- Editor:** Displays the 'LoginTest.feature' file with the following content:

```
1 Feature: Failed Test Case
2
3 Scenario: Login With Invalid UserName
4   Given User should open Chrome Browser
5   When User should Enter url in Browser
6   When User should Navigate to Home Page
7   And Enter Invalid UserName and Valid Password in Edit Box
8   And Click On Login PushButton
9   Then Error Message should be displayed
10  And Verify Error Message Text
```
- Test Results:** Shows the test execution summary:
 - Finished after 7.679
 - Runs: 7/7
 - Errors: 0
 - Failures: 0A green progress bar indicates successful completion. Below this, a 'Failed Test Case [Runner: Cucumber] 7.679s' entry is listed with a 'Failure Trace' link.
- Console:** Displays the execution log:

```
<terminated> TestCase2.feature [Cucumber-JVM] C:\Program Files\openjdk-21.0.2\bin\javaw.exe (22-Jun-2026, 10:46:33 am - 10:46:43 am elapsed 0:00:10) [pid: 15424]
Jun 22, 2026 10:46:38 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 149, returning the closest version; found: 148; Please update to a Selenium version that supports CDP
Error: Your username is invalid!

1 scenarios (1 passed)
7 steps (7 passed)
0m 9.102s
```

Cucumber - cucumber.first.example/Futures/LoginTest.feature - Eclipse IDE

File Edit Navigate Search Project Run Window Help

Package Explorer

- cucumber.first.example
 - src/main/java
 - src/main/resources
 - src/test/java
 - src/test/resources
 - JRE System Library [jdk-21.0.2]
 - Maven Dependencies
 - Futures
 - LoginTest.feature
 - TestCase2.feature
 - testCase3.feature
 - src
 - target
 - pom.xml

LoginTest.feature

```
1 Feature: Checking login Functionality
2
3 Scenario: Successful Login with Valid Credentials
4   Given User should open Chrome Browser
5   When User should Enter url in Browser
6   When User should Navigate Home Page
7   And Enter UserName and Password in Edit Box
8   And Click On Login PushButton
9   Then Message displayed Login Successfully
```

Problems Javadoc Declaration Test Results

Finished after 8.388

Runs: 6/6 Errors: 0 Failures: 0

Checking login Functionality [Runner: Cucumbe] Failure Trace

Console

```
<terminated> LoginTest.feature [Cucumber-JVM] C:\Program Files\openjdk-21.0.2\jdk-21.0.2\bin\javaw.exe (22-Jun-2026, 10:46:03 am - 10:46:14 am elapsed 0:00:11) [pid: 26484]
Jun 22, 2026 10:46:08 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 149, returning the closest version; found: 148; Please update to a Selenium version that supports CDP
Navigated to login page
Login Success URL: https://practicetestautomation.com/logged-in-successfully/

1 scenarios (1 passed)
6 steps (6 passed)
0m 9.822s
```

31°C Mostly sunny 10:46 AM 22-06-2026

Git Repositories

Module Name: GitHub Repositories

No. of Repositories:

Emp ID: 85009810

Submitted By : Ananya Pattanaik

Submitted To: Raghavendra BN

S.No	Date	Project Name	Repository Link
1	03/06/2026 - 05/06/2026	Java Programs	https://github.com/ananyapattanaik/Basic-Java-Programs
2	08-06-2026	SeleniumWebDriver-BrowserTesting	https://github.com/ananyapattanaik/JavaEclipseProgram-LaunchBrowserTestCase
3	08-06-2026	TestNG-BrowserTesting, SeleniumWebDriver-CaseStudy1-BrowserTesting	https://github.com/ananyapattanaik/JavaEclipseProgram-LaunchBrowserTestCase
4	09-06-2026	SeleniumWebDriver -CaseStudy -Commands,SeleniumWebDriver -CaseStudy_Locaters-EasyCalculation	https://github.com/ananyapattanaik/SeleniumWebdriver-Locators-NavigationCommands
5	10-06-2026	SeleniumWebBrowser _Case Study-DropDownList,Radio Button,ScrollUpDown, capturing the links and image	https://github.com/ananyapattanaik/SeleniumWebdriver-Locators-NavigationCommands
6	11-06-2026	SeleniumWebDriver _CaseStudy-POM-Practicetestautomation,,	https://github.com/ananyapattanaik/MavenProject
7	11-06-2026	SeleniumWebDriver _CaseStudy-POM-Profilebrightest website	https://github.com/ananyapattanaik/MavenProject
8	11-06-2026	SeleniumWebDriver _CaseStudy-POM- EasyCalculation Website,	https://github.com/ananyapattanaik/MavenProject
9	12-06-2026	DataDrivenTesting-ApachePOI	https://github.com/ananyapattanaik/DataDrivenTesting-ApachePOI
10	12-06-2026	Hybrid driven FrameWork	https://github.com/ananyapattanaik/HybridDrivenFramework
11	13-06-2026	Synchronization-Selenium	https://github.com/ananyapattanaik/Selenium-Synchronization
12	13-06-2026	Multiple Browser Testing	https://github.com/ananyapattanaik/MultipleBrowserTesting
13	13-06-2026	Assertion-TestNG	https://github.com/ananyapattanaik/Assertion-TestNG
14	13-06-2026	DataProvider-TestNG	https://github.com/ananyapattanaik/TestNG-DataProvider
15	14-06-2026	Assignment 1	https://github.com/ananyapattanaik/SeleniumJavaAssignment1
16	14-06-2026	Assignment 2	https://github.com/ananyapattanaik/Assignment-2
17	14-06-2026	Assignment 6	https://github.com/ananyapattanaik/Assignment-6
18	15-06-2026	Assignment 7	https://github.com/ananyapattanaik/Assignment-7
19	15-06-2026	Assignment 8	https://github.com/ananyapattanaik/Assignment-8
20	15-06-2026	Assignment 9	https://github.com/ananyapattanaik/Assignment-9
21	16-06-2026	Assignment 10	https://github.com/ananyapattanaik/Assignment-10
22	16-06-2026	Assignment 11	https://github.com/ananyapattanaik/Assignment-11
23	16-06-2026	Assignment 12	https://github.com/ananyapattanaik/Assignment-12
24	16-06-2026	Assignment 13	https://github.com/ananyapattanaik/Assignment-13
25	16-06-2026	Assignment 14	https://github.com/ananyapattanaik/Assignment-14
26	17-06-2026	Cucumber-TestCases	https://github.com/ananyapattanaik/Cucumber-TestCases
27	19-06-2026	RestAssured TestCases	https://github.com/ananyapattanaik/RestAssured-TestCases



Thank You